

LAB 6: Unsupervised Learning

OBJECTIVE: Perform clustering using K-Means from scratch.

DATASET: Iris Dataset

THEORY:

- Unsupervised learning finds patterns in data without labels.
- K-Means clustering partitions data into k clusters by:
 - Randomly initializing k centroids
 - Assigning each point to the nearest centroid
 - Updating centroids based on the mean of points in each cluster
 - Repeating until convergence
- Inertia measures how close data points are to their assigned cluster centers (lower is better).
- Tools like Scikit-learn simplify clustering and visualization.

ALGORITHM (K-MEANS CLUSTERING):

- Load dataset (Iris or synthetic)
- Select the number of clusters k
- Initialize centroids randomly
- Repeat until convergence:
 - Assign each data point to the nearest centroid
 - Update centroids as the mean of assigned points
- Compute inertia (sum of squared distances)
- Visualize clusters and centroids

SOURCE CODE:

CELL 1: IMPORT LIBRARIES

```
In [13]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
```

CELL 2: LOAD DATASET

```
In [14]: # Load Iris dataset
iris = load_iris()
X = iris.data[:, :2] # use first two features for easy 2D visualization

print("Features shape:", X.shape)
```

Features shape: (150, 2)

CELL 3: APPLY K-MEANS

```
In [15]: # Define number of clusters
k = 3

# Initialize and fit KMeans
kmeans = KMeans(n_clusters=k, random_state=42)
kmeans.fit(X)

# Get cluster labels and centroids
labels = kmeans.labels_
centroids = kmeans.cluster_centers_

# Inertia
print("Inertia (sum of squared distances):", kmeans.inertia_)
```

Inertia (sum of squared distances): 37.05070212765958

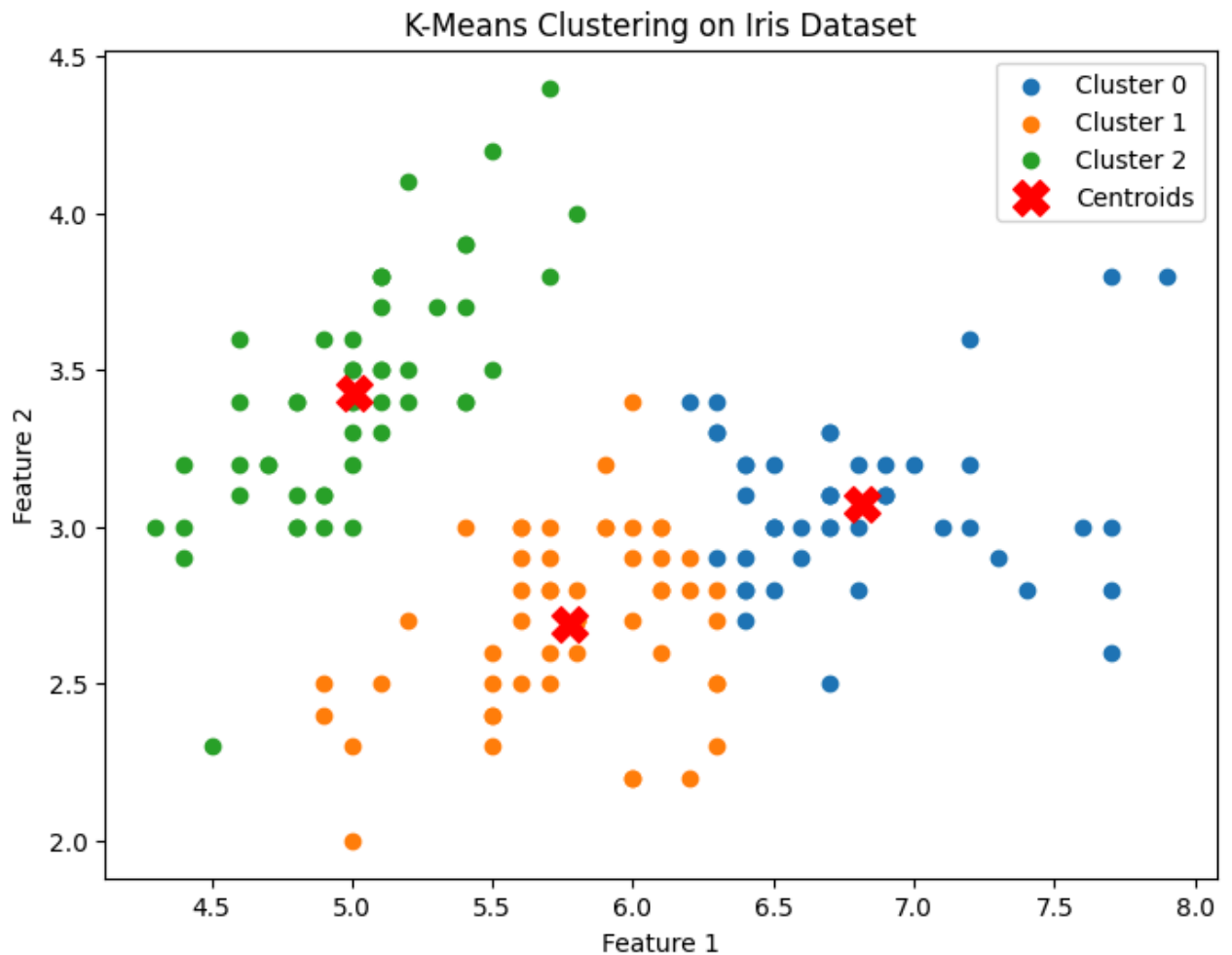
CELL 4: VISUALIZE CLUSTERS

```
In [16]: plt.figure(figsize=(8,6))

# Plot clusters
for i in range(k):
    plt.scatter(X[labels==i, 0], X[labels==i, 1], label=f'Cluster {i}')

# Plot centroids
plt.scatter(centroids[:,0], centroids[:,1], s=200, c='red', marker='X', label='Centroids')

plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("K-Means Clustering on Iris Dataset")
plt.legend()
plt.show()
```



RESULTS:

- K-Means clustered the Iris dataset into 3 distinct groups, mostly aligning with actual species.
- Cluster centroids are clearly visible on the plot.
- Inertia (sum of squared distances) was low, indicating compact clusters.
- Visualization confirms that similar points are grouped together effectively.

CONCLUSION:

- K-Means is effective for unsupervised clustering when the number of clusters is known.
- The algorithm groups similar data points without using labels.
- Inertia provides a quantitative measure of cluster compactness, while visualization helps interpret the clusters.
- This lab demonstrates the ability of unsupervised learning to uncover patterns in unlabeled data.